

188422

Subject : Basic Electricity

M.M. : 50

Note: Multiple Choice questions. All questions are compulsory. (5x1=5)

a) henry b) ohm
c) farad d) tesla

a) capacitor b) inductor
c) resistor d) transformer

- a) To store energy
- b) to measure voltage
- c) to protect circuits from over current
- d) to convert ac to dc

Q.4 What happens when a lead -acid battery is discharged?

- a) the voltage increases
- b) the electrolyte turns to water
- c) the plates become clean
- d) it generates hydrogen gas

Q.5 What does the power factor indicate in an AC circuit?

- a) the efficiency of power usage
- b) the amount of resistance in the circuit
- c) the strength of the magnetic field
- d) the amplitude of the voltage

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

Q.6 Define the term “Power factor” in AC circuits.

Q.7 What is the function of an inductor in an AC circuit?

Q.8 Name two different types of capacitors.

Q.9 What is the difference between a primary and secondary cell?

(2)

188422

Q.10 What are the components of an RLC series circuit?

SECTION-C

Note: Short answer type questions. Attempt any six questions out of Eight questions. (6x5=30)

Q.11 What is the role of a fuse in an electrical circuit?

Q.12 Explain the principle of an electric generator.

Q.13 Describe the applications of lead-acid batteries.

Q.14 What is the difference between peak value and RMS value in AC circuits?

Q.15 Explain the importance of Faraday’s Laws in electrical engineering.

Q.16 What is the significance of the phase difference in AC circuits?

Q.17 Describe how an electromagnetic field is created in a solenoid.

Q.18 How does the crest factor affect AC waveforms?

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Explain the process of AC voltage generation and describe its waveform.

Q.20 Discuss in detail the construction, working, and applications of capacitors in electrical circuits.

(3)

188422